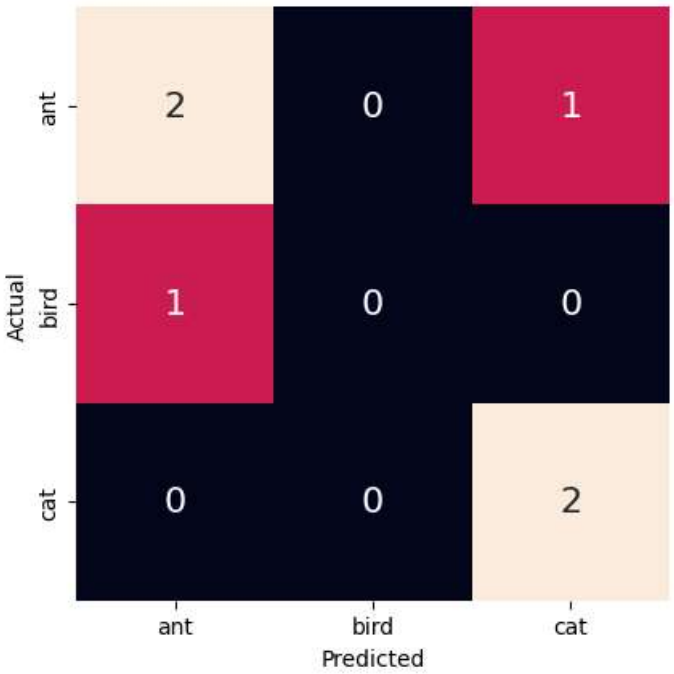
```
import numpy as np
import seaborn as sn
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
```

```
y_true = ["cat", "ant", "cat", "cat", "ant", "bird"]
y_pred = ["ant", "ant", "cat", "cat", "ant", "cat"]
labels=["ant", "bird", "cat"]
cm = confusion_matrix(y_true, y_pred, labels=labels)

# Adjust the confusion matrix to show TP in the top right corner
cm = np.flip(cm)
```

```
df_cm = pd.DataFrame(cm, labels, labels)
ax = sn.heatmap(df_cm, annot=True, annot_kws={"size": 16}, square=True, cbar=False, fmt='g')
ax.set_ylim(0, 3) #this manually corrects the cutoff issue in sns.heatmap found in matplotlib ver 3.1.1
plt.xlabel("Predicted")
plt.ylabel("Actual")
ax.invert_yaxis() #optional
plt.show()
```



Y_true = [cat, ant, cat, cat, ant, bird]

Y_pred = [ant, ant, cat, cat, ant, cat]

Y_true = [3, 1, 3, 3, 1, 2] → # of classes

Y_Pred = [1, 1, 3, 3, 1, 3]

Ant 0
Bnd 1
Cat 2

class-1

	T	F
T	TP 2	FN 1
F	FP 0	TN 3

2

class-2

	T	F
T	0	1
F	0	3

1

class-3

	T	F
T	2	1
F	1	2

3

Actual \ Pred	1	2	3
1	2	0	0
2	0	0	1
3	1	0	2

Multi-class table

Confusion Matrix

Actual \ 1 Predicted	1	0
1	TP	
0		TN

Colum false positive

row false negative

Class 1	class 2	Class 3
$P = \frac{2}{2} = 1$	$\frac{0}{0} = 0$	$\frac{2}{3} = 0.67$
$R = \frac{2}{3} = 0.67$	$\frac{0}{1} = 0$	$\frac{2}{3} = 0.67$
$F1 = \frac{2 \times 0.67 \times 1}{1.67} = 0.80$	0	$F1 = \frac{2 \times 0.67 \times 0.67}{0.67 + 0.67} = 0.67$

Macro_F1 → $\frac{0.80 + 0 + 0.67}{3} = 0.49$ | Weight F1: $\frac{(2 \times 0.8 + 1 \times 0 + 3 \times 0.67)}{6}$ $\boxed{0.66}$

```
print('\nClassification Report\n')
print(classification_report(y_true, y_pred))
```

```
Classification Report

              precision    recall  f1-score   support

         ant       0.67      1.00      0.80         2
        bird       0.00      0.00      0.00         1
         cat       0.67      0.67      0.67         3

    accuracy                           0.67         6
   macro avg       0.44      0.56      0.49         6
weighted avg       0.56      0.67      0.60         6

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning: Pre
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning: Pre
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning: Pre
  _warn_prf(average, modifier, msg_start, len(result))
```